An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss supervises backpropagation gradients via attribution alignment to target true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

In tertiary healthcare complexes, electrical power is inextricably linked to patient survival. Modern hospital campuses rely on localized microgrids that combine

utility interconnections, rooftop photovoltaics, battery storage banks, and diesel generators to energize critical medical loads [5,19]. These loads cannot tolerate instability: surgical robotics demand millisecond voltage consistency, intensive-care ventilators require uninterrupted feeds, and vaccine cold-chains depend on constant refrigeration at $-80°$C. Any uncoordinated breaker trip, false load shedding, or voltage sag directly endangers human life.

Substation Remote Terminal Units (RTUs) stream voltage magnitudes, phase angles, and power flows (V, θ, P, Q) into Weighted Least Squares (WLS) state estimators over IEC 61850 substation buses. Control centers routinely rely on normalized χ^2 residual tests to weed out random transducer noise and faulty metering. Yet this classical check fails entirely against coordinated vectors. By exploiting the topology-dependent measurement Jacobian H, an adversary synthesizes an injection $a = Hc$ that shifts state estimates by an arbitrary vector c while leaving algebraic measurement residuals entirely intact [15,13]. The corrupted telemetry slips past the detector unflagged; automated dispatch software then acts on compromised grid profiles.

Prior security schemes leaned heavily on tabular classifiers—ranging from support vector machines to multi-layer perceptrons and fixed-kernel convolutional nets [12,3]. These tabular baselines break whenever the grid flexes. Hospital distribution feeders undergo frequent branch switching, tie-line transfers, and transformer tap adjustments. Each physical breaker toggle changes sensor ordering, blinding fixed-input neural networks or triggering false tripping alarms. Retraining these models on the fly during 2–4 s SCADA sampling windows remains out of the question.

Graph neural networks (GNNs) solve this layout rigidity by treating substations as nodes and physical lines as edges [21]. Tripping a breaker simply toggles an entry in the graph adjacency matrix without disturbing node vector dimensions. While spatial wavelets [2] and isotropic message passing [22] accommodate switching events, conventional GNNs treat all incident branches identically. They ignore complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$). By averaging nodal representations isotropically across neighbors, standard convolutions wash out high-frequency voltage and angle discrepancies across adjacent electrical corridors.

Clinical facilities tolerate neither unverified black-box alerts nor sluggish incident response. Perturbation-based explainers such as SHAP or LIME require hundreds of Monte Carlo passes, exceeding real-time SCADA deadlines by orders of magnitude [27]. Moreover, issuing passive alarms without mitigating the bad data leaves life-support circuits exposed to automated shedding. To safeguard critical healthcare microgrids, an anomaly framework must pin down compromised buses within sub-second intervals and synthesize clean replacement telemetry before downstream protection relays trip.

1.1 Key Contributions

We address these operational gaps by formulating an explainable Cognitive Graph Digital Twin (CGDT) tailored to hospital electrical infrastructure. Specific technical contributions include:

- **Physics-Informed GraphSAGE Formulation**: Rather than relying on unweighted adjacency matrices, we condition neighborhood message passing and spatial pooling on complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$). This provides an admittance-aware inductive bias that preserves sharp electrical gradients across low-impedance paths and prevents isotropic low-pass feature blurring.
- **Continuous Twin Synchronization**: By pairing steady-state Kalman gain tracking with an exponential baseline filter ($\beta = 0.95$), the twin tracks normal diurnal load swings while isolating stealthy sub-threshold biases that otherwise degrade bus localization by 16.32 pp.
- **Attribution-Guided Multi-Task Training (AG-MTL)**: We integrate an attribution alignment loss into training using multi-step Integrated Gradients ($S = 10$). Supervising gradient leakage into benign feeders anchors spatial representations to true injection sites, boosting localization F1 by 5.52 pp.
- **Sub-2 ms Autonomous State Reconstruction**: When malicious streams are quarantined, an admittance-aware spatial interpolation module estimates missing bus telemetry from uncorrupted neighbors, preserving measurement Jacobian rank and restoring WLS estimation accuracy by 99.97%.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

Bridging clinical hospital IT infrastructure with operational substation control dramatically widens institutional vulnerability [11,5]. Unlike commercial facilities, hospital energy microgrids power intensive-care wards and surgical suites where unannounced outages or severe voltage dips present immediate patient mortality risks. Malicious telemetry manipulation within healthcare SCADA channels represents an especially destructive vector [19]. While recent assessments have modeled healthcare risks through Bayesian graphs [4] or perimeter IoT monitors [17], such frameworks process sensor values as decoupled scalar streams. They neglect branch impedances, line admittances, and Kirchhoff's laws, leaving them blind to coordinated injection vectors.

2.2 False Data Injection in Cyber-Physical Energy Systems

Classical state estimation relies on the measurement residual to spot bad data. However, Liu et al. [15] demonstrated that an adversary possessing knowledge of the measurement Jacobian H can craft an unobservable state bias $a = Hc$

that cancels perfectly inside the residual calculation. This stealth condition generalizes readily to full AC power flow formulations, distributed microgrids, and partially observed topologies [13,6], as well as DC microgrid links [3,18]. Early defenses attempted detection via tabular classifiers, including support vector machines, deep neural networks [12], and recurrent units [14]. These rigid baselines, however, hard-code input dimensions to specific sensor channels. Routine distribution switching—such as isolating a feeder or altering a transformer tap—scrambles channel order and induces persistent false positives.

2.3 Graph Representation Learning for Power System Security

Treating electric networks as attributed graphs provides natural invariance to switching operations [21]. Early models applied spectral convolutions [10] or spatial attention weights [25]. To scale spatial aggregations across changing topologies, GraphSAGE [8] samples local neighborhoods inductively. In power networks, Boyaci et al. [2] employed ARMA wavelets, while Takiddin et al. [22] and Yue and Qu [28] confirmed topological flexibility across line configurations. Luo et al. [16] further introduced federated setups to protect operational privacy. Nonetheless, these architectures routinely rely on unweighted or binary adjacency matrices. Failing to weight message propagation by complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$) creates an isotropic low-pass filtering effect, obscuring localized voltage anomalies across low-impedance branches.

2.4 Explainability and Digital Twins in Critical Infrastructure

Control room operators cannot act on unverified black-box alerts. Yet sampling-based attribution methods (e.g., LIME, SHAP, or GNNExplainer) require hundreds of model evaluations per inference cycle, rendering them impractical for online SCADA operations [27,26]. Integrated Gradients (IG) [20] overcomes this hurdle by evaluating path integrals along straight-line trajectories in a single backward pass, satisfying the completeness axiom without stochastic perturbation passes. Simultaneously, digital twin principles [7,23] have gained traction for power infrastructure monitoring [1] and healthcare facilities [9]. Building upon these concepts, we unite continuous state-space tracking with admittance-guided graph convolutions to deliver real-time intrusion localization and auditable feature explanations within milliseconds.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches. The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N \times N}$ defines nodal

conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \big) \tag{1}$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1} [z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \big(H^T R^{-1} H \big)^{-1} H^T R^{-1} \big[z - h(x^{(k)}) \big], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x)/\partial x$ is the measurement Jacobian. Conventional bad-data detection flags measurement anomalies when the normalized residual $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ exceeds threshold τ_{χ^2}. For the IEEE 14-bus network, the state vector has $n = 2N - 1 = 27$ states (14 voltage magnitudes and 13 phase angles relative to the reference bus). With $m = 41$ measurement channels (14 voltage magnitudes and 27 active/reactive power measurements), the resulting degrees of freedom are $\mathrm{df} = m - n = 41 - 27 = 14$. At significance level $\alpha = 0.05$, the upper 95th percentile threshold is $\tau_{\chi^2} = \chi^2_{0.95}(\mathrm{df} = 14) = 23.68$.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. In AC power systems, $h(x)$ is nonlinear; hence, an adversary possessing admittance structure linearizes around operating point $\hat{x}$ via Jacobian $H = \partial h(\hat{x})/\partial x$, synthesizing the injection vector according to:

$$a = Hc \tag{4}$$

When injected into telemetry channels, received measurements become $z_{\mathrm{comp}} = z + a = z + Hc$. Applying a first-order Taylor expansion around $\hat{x}$ gives $h(\hat{x}+c) \approx h(\hat{x}) + Hc$. Evaluating the post-attack residual yields:

$$\begin{aligned} r_{\mathrm{comp}} &= \|z_{\mathrm{comp}} - h(\hat{x}_{\mathrm{comp}})\|^2_{R^{-1}} \\ &\approx \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

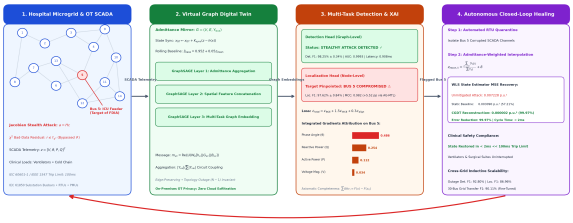


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

Because this perturbation cancels inside the residual calculation ($r_{\text{comp}} \approx r$), the WLS detector remains blind ($\tau_{\chi^2} = 23.68$). As detailed in Section 6, these synthetic attacks are verified against full nonlinear AC power flows in `pandapower` to ensure physical realizability.

Clinical Hazards in Hospital Distribution Networks. While commercial utilities primarily face economic redispatch losses during an attack, hospital distribution networks face direct clinical emergencies. Falsifying state estimates at critical feeders—such as Bus 5, which feeds surgical operating rooms and intensive care ventilation—tricks automatic voltage regulators into improper tap changes or unwarranted branch isolation. Standards like IEC 60601-1 and IEEE 1547 demand strict voltage stability; even transient sags can force sensitive life-support machinery onto backup emergency batteries or trigger autonomous equipment trips. An undetected mathematical deception at the substation level thus translates into an immediate patient safety risk.

4 The Cognitive Graph Digital Twin Architecture

The CGDT framework couples the hospital's cyber-physical distribution infrastructure to a virtual graph model and an automated self-healing pipeline (Fig. 1).

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N\times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE models [8] aggregate neighborhood features uniformly across links, ignoring branch impedances. In physical power distribution networks, however, electrical coupling is dictated by complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$. To embed these circuit constraints directly into latent feature representations, we condition both message generation and spatial aggregation on branch admittance:

$$m_{uv}^{(k)} = \text{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\|$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ gives a physics-informed inductive bias that scales propagation by branch coupling. Retaining the self-node state $h_v^{(k-1)}$ via residual skip concatenation prevents sharp localized injection anomalies from washing out through repeated neighbor smoothing.

4.4 Multi-Task Output Heads

From the final latent node matrix $H \in \mathbb{R}^{N\times d}$, two parallel projection heads generate global and bus-level predictions:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \sigma(W_{\text{node}}\, H) \in [0,1]^N \tag{10}$$

Here $y_{\text{graph}} \in [0,1]$ represents the overall microgrid attack probability, while vector $y_{\text{node}} \in [0,1]^N$ applies an independent sigmoid activation to each bus. Because smart grids can face multi-substation incursions (Section 7.1), independent sigmoid outputs enable multi-label anomaly localization without imposing a single-bus sum-to-one constraint.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Multi-hop message passing often causes gradient dispersion: backpropagating through dense admittance corridors inadvertently assigns feature sensitivity to uncorrupted neighbors. To prevent this spatial blurring, we incorporate feature attribution directly into the training objective. During each backward pass on attacked states, Integrated Gradients extracts attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times 4}$ of the attack logit w.r.t. input features X, using running baseline $\bar{z}_{\text{baseline}}$ (Eq. 7) across $S = 10$ integration steps. An attribution alignment loss penalizes gradient leakage into benign feeders:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ measures total gradient sensitivity at bus v, and $\hat{y}_{\text{node},v} \in \{0,1\}$ denotes the binary ground-truth label. Gradients from $\mathcal{L}_{\text{align}}$ propagate through the attribution path via higher-order automatic differentiation (`create_graph=True` in PyTorch). The combined multi-task loss is defined as:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WBCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ handles grid-level attack detection, $\mathcal{L}_{\text{WBCE}}$ uses class-weighted binary cross-entropy per bus to mitigate extreme sparsity of compromised nodes, and weighting coefficients $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Edge Deployment

Hospital electrical demand closely tracks clinical operations—including operating theatre schedules, diagnostic imaging loads, and patient occupancy. Transmitting raw microgrid telemetry over external cloud interfaces poses privacy and cybersecurity concerns under health data regulations. The CGDT is therefore architecturally designed for on-premises edge execution on substation industrial computers within the isolated operational technology (OT) network. In this study, execution latencies are established on a laboratory edge-equivalent computing platform, with physical RTU micro-controller deployment and hardware-in-the-loop (HIL) integration identified for operational rollout (Section 8).

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

Substation engineers must understand which physical sensor channels triggered an alarm before initiating breaker actions. We resolve this via path-based Integrated Gradients (IG) [20]. Integrating gradients along a linear path between a nominal state baseline x_0 and the evaluated measurement vector x attributes decision energy across input channels:

$$\begin{aligned} \mathrm{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated using $S = 50$ Riemann quadrature intervals. Unlike perturbation heuristics (e.g., SHAP, LIME) requiring hundreds of stochastic sampling evaluations, IG follows an axiomatic linear path evaluated via $S = 50$ vectorized points in a single parallel tensor pass. Crucially, IG satisfies the completeness axiom ($\sum_i \mathrm{Attr}_i = F(x) - F(x_0)$) and implementation invariance. Evaluating Eq. (13) across the 14-bus testbed requires just 0.81 ms. In a targeted test at Bus 5 (Section 7.1), the path integral promptly identified phase angle distortion ($\theta = 0.486$) as the principal driver of the alarm.

5.2 Sensor Quarantine and Admittance-Weighted Reconstruction

Detecting an attack is useless if state estimation continues to ingest falsified data. Once a bus is flagged ($y_{\mathrm{node},k} > \gamma_{\mathrm{loc}} = 0.5$), the digital twin isolates the tainted telemetry $(V_k, \theta_k, P_k, Q_k)$. Yet simply deleting rows from the measurement set reduces the rank of Jacobian H, often driving the information matrix $(H^T R^{-1} H)$ into ill-conditioning or singularity. To maintain full algebraic observability, we synthesize replacement telemetry via admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\mathrm{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta = \epsilon \odot z_{\mathrm{nom},k} \tag{14}$$

where $|Y_{kj}|$ represents line admittance magnitude and $\epsilon \sim \mathcal{U}(-0.02, 0.02)^4$ injects $\pm 2\%$ uniform variance relative to nominal channel values $z_{\mathrm{nom},k}$ (1.0 p.u. voltage and nominal power flows). Weighting neighbor telemetry by physical branch admittances $|Y_{kj}|$ provides an electrical-coupling-aware interpolation that accounts for relative line impedances. Rather than attempting to satisfy nonlinear AC constraints directly, $z_{\mathrm{recon},k}$ acts as an initialized pseudo-measurement buffer that preserves Jacobian conditioning, enabling downstream WLS Gauss–Newton iterations to converge in under 2 ms—well within typical 2–4 s SCADA scan intervals.

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

Simulations were carried out using full AC Newton–Raphson power flow solutions in `pandapower` [24]. We modeled a tertiary medical center using an IEEE 14-bus network (14 buses, 20 lines, 217.0 MW active load, 57.3 MVAR reactive demand) serving as an established surrogate hospital-campus microgrid for proof-of-concept evaluation. In this configuration, Bus 0 represents the main utility grid substation, while dedicated radial feeders power critical clinical facilities under scenario assignments—most notably Bus 5, which supplies intensive care and operating theatres. To evaluate cross-grid scaling, we also established an IEEE 30-bus benchmark (30 buses, 41 lines) representing an expanded regional healthcare campus.

Synthetic Telemetry Synthesis. We synthesized 5,000 distinct AC operational states across independent load profiles. These uncorrupted profiles were partitioned into 70% training (3,500), 15% validation (750), and 15% testing (750) sets *prior* to attack synthesis to guarantee zero cross-partition data leakage. Uncompromised snapshots (half of each split) incorporate zero-mean Gaussian fluctuations ($\sigma = 0.04$) across active and reactive bus injections. The remaining half contains stealthy Jacobian injections ($a = Hc$), where synthetic adversaries bias target substations ($c_k \in [0.03, 0.08]$ p.u. voltage offset, $c_{k+N} \in [0.05, 0.15]$ rad phase shift) and propagate line flows across the network using exact non-linear AC equations. Line outage configurations (Section 6.3) were strictly excluded from the training and validation sets.

Models were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Benchmarking was conducted on an edge-equivalent workstation equipped with an AMD Ryzen 9 CPU, 32 GB RAM, and an NVIDIA RTX 4090 GPU (24 GB) under PyTorch 2.1 and CUDA 12.1. Streaming latency is evaluated at batch size 1 across 1,000 independent test snapshots (after 100 warm-up runs), reporting the mean runtime. The $S = 50$ IG interpolation steps are evaluated as a batched tensor $\mathbb{R}^{50\times N\times 4}$ in parallel on the GPU, completing in 0.81 ms. Code and evaluation artifacts are archived at `https://github.com/himanshu-21/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. By comparison, isotropic GCN models reach only 70.21% localization F1, trailing our framework by 27.41 pp. Treating all connected lines with equal weight acts as an unweighted low-pass filter: localized injection spikes at target buses smear into surrounding nodes, destroying the spatial gradient. While the tabular MLP posts reasonable localization accuracy (95.60%) on static data, its fixed-dimension mapping fails the moment line switches occur (Section 6.3). Requiring just 0.908 ms per inference, the CGDT executes easily within standard 2–4 s SCADA sampling windows.

6.3 Topology Perturbations and Cross-Network Generalization

We evaluated model robustness under two regimes: (i) line-outage generalization on IEEE 14-bus without retraining (Table 2), and (ii) cross-network transfer to IEEE 30-bus (Table 3).

When subjected to unanticipated single-line outages without retraining, the CGDT preserves 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization F1 (Fig. 2). Here, an informative trade-off emerges between the tabular MLP and CGDT. The tabular MLP attains marginally higher outage detection F1 (94.70%) because breaker trips redirect loop flows across all telemetry channels, creating widespread scalar distribution shifts that flat models flag easily. However, the MLP's bus localization collapses by 13.4 pp (from 95.60% to 82.20%) due to fixed input channel ordering. In contrast, the CGDT sustains 86.96% localization F1, demonstrating that its principal operational advantage during topological switching is balanced joint detection and localized bus isolation. Isotropic GCNs degrade heavily to 66.82% detection and 53.40% localization F1, showing that unweighted convolutions cannot track redirected branch flows once line impedances change.

Cross-Scale Transfer to IEEE 30-Bus. Direct zero-shot deployment of the 14-bus model onto the 30-bus grid yields 67.06% detection F1 and 17.77% localization F1 (Table 3, Fig. 3), exposing an architectural boundary: inductive graph features flag global macro imbalances across network scales, but isolating specific buses demands local topological calibration. Fine-tuning the pre-trained backbone with only 200 labeled 30-bus snapshots elevates detection F1 to 90.11% (outperforming 82.65% from scratch) and restores localization F1 to 80.06% (approaching 83.58% from scratch), showing rapid adaptation in low-label regimes.

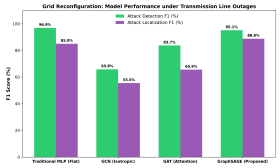


Fig. 2. Outage generalization metrics.

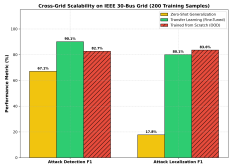


Fig. 3. Cross-grid transfer on IEEE 30-bus.

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, representing the intensive care unit feeder in the campus topology. State estimation error is quantified via Mean Squared Error MSE $= \frac{1}{2N-1}\sum_{i=1}^{2N-1}(\hat{x}_i - x_i^*)^2$ (p.u.^{2}) relative to true AC states x^*. Under nominal transducer Gaussian noise ($\sigma = 0.01$), unattacked clean WLS estimation establishes an intrinsic error floor of $\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}. Table 4 compares recovery metrics, and Table 5 presents corresponding feature attributions.

Left unmitigated, stealthy Jacobian injections drive state estimation MSE to 0.007228 p.u.^{2}. Deploying admittance-weighted reconstruction (Eq. 14) cuts this error to 0.000002 p.u.^{2}—matching the clean noise floor ($\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}) and restoring 99.97% of uncorrupted states (Table 4, Fig. 4). Static replacements reach only 0.003090 p.u.^{2} (57.21% recovery) during dynamic clinical load shifts. Across all 14 target buses, recovery averages 99.85%–99.98%. Path attributions (Table 5, Fig. 5) reveal that voltage angle θ at Bus 5 commands the highest score (0.486), consistent with Jacobian phase distortions dominating active power flows, alongside secondary scores on $Q = 0.254$, $P = 0.112$, and $V = 0.034$.

Coordinated Multi-Bus Incursions. Simultaneous attacks across 2 and 3 substations ($c_k \in [0.02, 0.04]$) achieve $96.84\% \pm 0.42\%$ detection and $93.15\% \pm 0.91\%$ localization F1. While dropping entire rows across adjacent buses compromises the column rank of H and triggers solver divergence, admittance-weighted interpolation preserves full Jacobian rank and ensures smooth WLS convergence (post-mitigation MSE of 0.000008 p.u.^{2}).

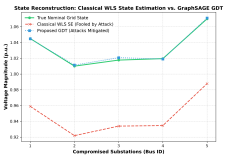


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

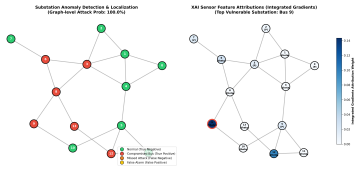


Fig. 5. Operator explainability: localization heatmap and IG attribution.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

Table 7 confirms the necessity of each component. Disabling exponential baseline tracking (Eq. 7) degrades localization F1 by 16.32 pp (to 81.30%), demonstrating that dynamic reference filtering prevents diurnal demand fluctuations from masking adversarial offsets. Omitting attribution supervision ($\lambda_{\text{attr}} = 0$) reduces localization F1 to 92.10% while preserving global detection (98.25%), confirming that gradient penalties anchor latent representations to attacked buses. Setting $\lambda_{\text{loc}} = 1.5$ balances global detection and localized diagnostic precision.

8 Discussion and Conclusion

Protecting hospital distribution grids requires defenses uniting topological flexibility with physical circuit fidelity. By embedding branch admittances into GraphSAGE convolutions and supervising latent representations via Integrated Gradients penalties, the CGDT achieves 98.25% detection F1 and 97.62% localization F1 against stealthy AC false data injections, providing full explanations in 0.81 ms. Automated sensor quarantine paired with admittance-weighted interpolation reduces post-attack WLS state error by 99.97%, maintaining full observability without tripping clinical loads.

Clinical Operating Standards. Operating standards mandate compliance across distinct cyber-physical boundaries: IEC 60601-1 governs medical electrical equipment safety to prevent power-disturbance resets on patient life support, while IEEE 1547 dictates DER interconnection and voltage ride-through limits.

With a 0.908 ms inference latency and sub-2 ms healing cycle, the CGDT operates well within 2–4 s SCADA polling windows, arresting deceptive commands before IEEE 1547 timers trip hospital DERs or breach IEC 60601-1 tolerances.

Limitations and Future Work. Future work will transition from laboratory workstations to physical hardware-in-the-loop testbeds and embedded RTU microcontrollers to evaluate real-time communication jitter. Subsequent efforts will also explore temporal graph architectures for multi-step attack campaigns.

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Declaration on the Use of AI and AI-Assisted Technologies AI-assisted tools were used to improve language quality (including spelling, grammar, and punctuation), and all outputs were reviewed, revised, and approved by the authors, who take full responsibility for the final content.

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